

1.1.1 Failure Handling and Recovery

Purpose: To evaluate the framework's ability to detect a node failure, adapt its internal state tables accordingly, and converge to a new, correct routing topology. This is the most critical test of the protocol's dynamic functionality.

Validation Method: A node failure was simulated by deliberately powering off Node 5. The network was then observed for a period to allow the protocol's timeout and cleanup mechanisms to take effect. The PrettyTable dashboards and event logs of the remaining four nodes were inspected to track the sequence of updates and verify convergence to a stable, four-node topology.

Observations: The system responded correctly and predictably to the failure of Node 5. The process of detection and reconvergence unfolded in a clear, logical sequence across the remaining nodes:

1. **Neighbor Timeout:** After a few missed HELLO messages from Node 5 (specifically, after the NEIGHB_HOLD_TIME of 6 seconds expired), Node 3 correctly removed Node 5 from its Neighbor Table. This was the initial trigger for the network-wide reconvergence.
2. **MPR Selector Set Adaptation:** The failure of Node 5 initiated a multi-step adaptation centered around Node 3. First, as the direct link timed out, Node 5 was removed from Node 3's neighbor table. Concurrently, the rationale for Nodes 2 and 4 selecting Node 3 as an MPR—which was to cover their two-hop neighbor, Node 5—was nullified. As a result, both Node 2 and Node 4 correctly ceased to select Node 3 as an MPR in their subsequent state updates. The cumulative effect was that Node 3, having lost all its selectors, correctly ended up with an empty My MPR Selectors list. This demonstrates the protocol's ability to correctly retract selector status based on both direct link failures and changes in the broader two-hop neighborhood.
3. **Cascading MPR Re-calculation:** The changes in neighborhood information were processed during the next periodic MPR calculation cycle on each node. This led to a network-wide re-evaluation of MPR sets. Specifically, the nodes that previously selected Node 3 as an MPR reassessed their choices:
 - Node 2 and Node 4: Both nodes had previously selected Node 3 as an MPR for the sole purpose of covering their two-hop neighbor, Node 5. With Node 5 no longer reachable, this requirement vanished. During their next periodic recalculation, both Node 2 and Node 4 correctly determined that Node 3 was no longer necessary for two-hop coverage and updated their MPR Set accordingly.
 - Node 5: This node had also selected Node 3, but since it was no longer part of the network, its selection became irrelevant. This demonstrates that the periodic re-evaluation mechanism correctly adapts to changes in the two-hop neighborhood, even if those changes are discovered between calculation cycles.

4. Topology and Routing Table Convergence: The protocol correctly handled the propagation of these changes:
- The absence of HELLO messages from Node 5 led to its removal from all relevant 1- Hop and 2-Hop Neighbor tables.
 - The periodic `_cleanup_thread` on each node ensured that stale entries for Node 5 were purged from all Topology and Routing tables.
 - Within a few update cycles, all remaining nodes (Nodes 1, 2, 3, and 4) converged to a consistent state reflecting the new four-node topology. Their routing tables correctly showed no path to Node 5.

Figure 4.3 shows the new, smaller network topology after the failure, while Figures 4.4(a-d) show the final, converged state of the dashboards for the remaining nodes. These snapshots provide a comprehensive view of the new network state, confirming not only the removal of Node 5 from all routing tables but also the updated neighbor lists, the adapted MPR sets and selector lists, and the purged topology tables.

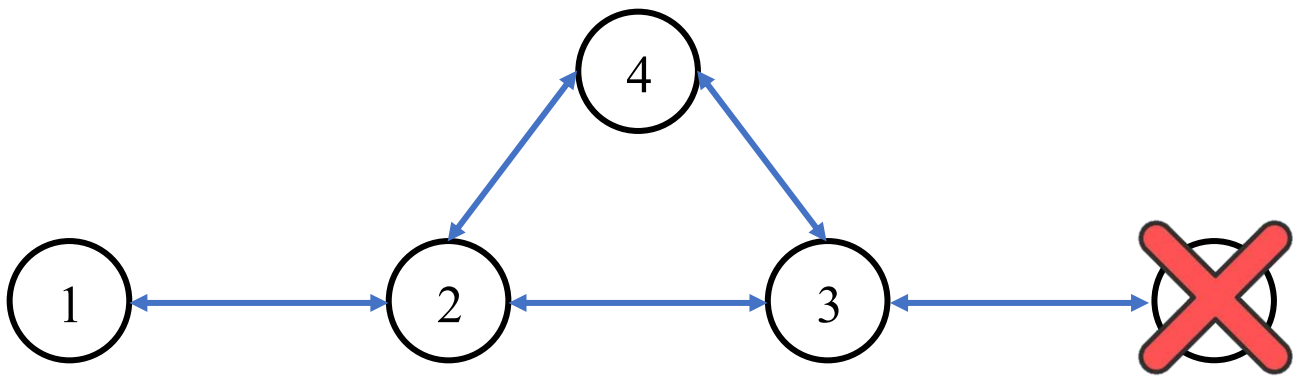


Figure 1: Converged Topology After Failure of Node 5

```
ras1@ras1: ~/Desktop
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--- Tables updated after HELLO from 192.168.10.2 (Node 192.168.10.1) ---
My MPR Set: ['192.168.10.2']
My MPR Selectors: (empty)
HNA Table (Gateways):
  (empty)
Interface Association Table (MID):
  - Interface: 10.10.10.2      | Main Addr: 192.168.10.2      (4.1s ago)
  - Interface: 10.20.20.2      | Main Addr: 192.168.10.2      (4.1s ago)
Neighbor Table (1-hop):
  - 192.168.10.2      | Status: SYM | Last Heard: 20:41:53 (0.0s ago)
2-Hop Neighbors Table:
  - 192.168.10.4      | via: 192.168.10.2      | Last Updated: 20:41:53 (0.0s ago)
  - 192.168.10.3      | via: 192.168.10.2      | Last Updated: 20:41:53 (0.0s ago)
Topology Table (Links):
  - (Dest: 192.168.10.1 , Last: 192.168.10.2 ) | Seq: 25 (4.1s ago)
  - (Dest: 192.168.10.3 , Last: 192.168.10.2 ) | Seq: 25 (4.1s ago)
  - (Dest: 192.168.10.4 , Last: 192.168.10.2 ) | Seq: 25 (4.1s ago)
Routing Table:
  - Dest: 192.168.10.2      | Next Hop: 192.168.10.2      | Hops: 1
  - Dest: 192.168.10.3      | Next Hop: 192.168.10.2      | Hops: 2
  - Dest: 192.168.10.4      | Next Hop: 192.168.10.2      | Hops: 2
Traffic Counts (Control): {'HELLO': 85, 'TC': 0, 'HNA': 34, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
```

Figure (a) Node State Dashboard for Node 1 After Node 5 Failure (192.168.10.1)

```
ras2@ras2: ~/Desktop
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My MPR Set: (empty)
My MPR Selectors: ['192.168.10.1', '192.168.10.3', '192.168.10.4']
HNA Table (Gateways):
  - Network: 0.0.0.0/0      | via Gateway: 192.168.10.1
Interface Association Table (MID):
  (empty)
Neighbor Table (1-hop):
  - 192.168.10.1      | Status: SYM | Last Heard: 20:41:52 (4.2s ago)
  - 192.168.10.3      | Status: SYM | Last Heard: 20:41:55 (1.5s ago)
  - 192.168.10.4      | Status: SYM | Last Heard: 20:41:57 (0.0s ago)
2-Hop Neighbors Table:
  (empty)
Topology Table (Links):
  - (Dest: 192.168.10.1 , Last: 192.168.10.2 ) | Seq: 26 (2.7s ago)
  - (Dest: 192.168.10.3 , Last: 192.168.10.2 ) | Seq: 26 (2.7s ago)
  - (Dest: 192.168.10.4 , Last: 192.168.10.2 ) | Seq: 26 (2.7s ago)
Routing Table:
  - Dest: 0.0.0.0/0      | Next Hop: 192.168.10.1      | Hops: 2
  - Dest: 192.168.10.1    | Next Hop: 192.168.10.1      | Hops: 1
  - Dest: 192.168.10.3    | Next Hop: 192.168.10.3      | Hops: 1
  - Dest: 192.168.10.4    | Next Hop: 192.168.10.4      | Hops: 1
Traffic Counts (Control): {'HELLO': 78, 'TC': 26, 'HNA': 0, 'MID': 31}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
```

Figure (b) Node State Dashboard for Node 2 After Node 5 Failure (192.168.10.2).

```
ras3@ras3: ~/Desktop
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My MPR Set: ['192.168.10.2']
My MPR Selectors: (empty)
HNA Table (Gateways):
- Network: 0.0.0.0/0 | via Gateway: 192.168.10.1
Interface Association Table (MID):
- Interface: 10.10.10.2 | Main Addr: 192.168.10.2 (9.1s ago)
- Interface: 10.20.20.2 | Main Addr: 192.168.10.2 (9.1s ago)
Neighbor Table (1-hop):
- 192.168.10.2 | Status: SYM | Last Heard: 20:41:57 (1.0s ago)
- 192.168.10.4 | Status: SYM | Last Heard: 20:41:55 (3.4s ago)
2-Hop Neighbors Table:
- 192.168.10.1 | via: 192.168.10.2 | Last Updated: 20:41:57 (1.0s ago)
Topology Table (Links):
- (Dest: 192.168.10.1 , Last: 192.168.10.2 ) | Seq: 26 (4.1s ago)
- (Dest: 192.168.10.3 , Last: 192.168.10.2 ) | Seq: 26 (4.1s ago)
- (Dest: 192.168.10.4 , Last: 192.168.10.2 ) | Seq: 26 (4.1s ago)
Routing Table:
- Dest: 0.0.0.0/0 | Next Hop: 192.168.10.2 | Hops: 3
- Dest: 192.168.10.1 | Next Hop: 192.168.10.2 | Hops: 2
- Dest: 192.168.10.2 | Next Hop: 192.168.10.2 | Hops: 1
- Dest: 192.168.10.4 | Next Hop: 192.168.10.4 | Hops: 1
Traffic Counts (Control): {'HELLO': 72, 'TC': 0, 'HNA': 0, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
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```

Figure 4.4 (c) Node State Dashboard for Node 3 After Node 5 Failure (192.168.10.3).

```
ras4@ras4: ~/Desktop
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--- MPR Set Recalculated (Node 192.168.10.4) ---
My MPR Set: ['192.168.10.2']
My MPR Selectors: (empty)
HNA Table (Gateways):
- Network: 0.0.0.0/0 | via Gateway: 192.168.10.1
Interface Association Table (MID):
- Interface: 10.10.10.2 | Main Addr: 192.168.10.2 (6.7s ago)
- Interface: 10.20.20.2 | Main Addr: 192.168.10.2 (6.7s ago)
Neighbor Table (1-hop):
- 192.168.10.3 | Status: SYM | Last Heard: 20:41:59 (1.5s ago)
- 192.168.10.2 | Status: SYM | Last Heard: 20:41:57 (3.5s ago)
2-Hop Neighbors Table:
- 192.168.10.1 | via: 192.168.10.2 | Last Updated: 20:41:57 (3.5s ago)
Topology Table (Links):
- (Dest: 192.168.10.1 , Last: 192.168.10.2 ) | Seq: 25 (11.7s ago)
- (Dest: 192.168.10.3 , Last: 192.168.10.2 ) | Seq: 25 (11.7s ago)
- (Dest: 192.168.10.4 , Last: 192.168.10.2 ) | Seq: 25 (11.7s ago)
Routing Table:
- Dest: 0.0.0.0/0 | Next Hop: 192.168.10.2 | Hops: 3
- Dest: 192.168.10.1 | Next Hop: 192.168.10.2 | Hops: 2
- Dest: 192.168.10.2 | Next Hop: 192.168.10.2 | Hops: 1
- Dest: 192.168.10.3 | Next Hop: 192.168.10.3 | Hops: 1
Traffic Counts (Control): {'HELLO': 64, 'TC': 0, 'HNA': 0, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
```

Figure (d) Node State Dashboard for Node 4 After Node 5 Failure (192.168.10.4).

